

Multiple Choice

1. **Answer:** C

Options: A) The organism is extinct in the wild and cannot be harvested for vaccine production; B) The organism that causes smallpox is too costly to use in vaccine production; C) The use of the actual virulent variola virus may be dangerous if too much is administered; D) Smallpox virus is confined to research laboratories; E) There is no specific treatment for smallpox

Note: Correct option: C - The use of the actual virulent variola virus may be dangerous if too much is administered

2. **Answer:** B

Options: A) Commensalism; B) Succession; C) Mutualism; D) Parasitism

Note: Correct option: B - Succession

3. **Answer:** A

Options: A) The body controls water excretion through sweat glands.; B) Water excretion is determined by the concentration of electrolytes in the body's cells.; C) The kidneys control water excretion by changing the size of the glomerulus.; D) Water excreted is controlled by the hormone insulin.; E) The body adjusts water excretion by altering the pH level of the blood.

Note: Correct option: A - The body controls water excretion through sweat glands.

4. **Answer:** D

Options: A) Consistent response to antibiotics across all protozoan forms; B) Inability to adapt to environmental conditions; C) Absence of common ancestral form; D) Extreme diversity, taxonomic organization, possession of certain plant-like features; E) Indistinct cellular structures among different protozoan groups

Note: Correct option: D - Extreme diversity, taxonomic organization, possession of certain plant-like features

5. **Answer:** B

Options: A) Cyanobacteria reduce sulfur compounds.; B) Cyanobacteria have no nuclei.; C) Green algae cannot reduce nitrogen compounds.; D) Green algae cannot photosynthesize.; E) Green algae produce cell membranes.

Note: Correct option: B - Cyanobacteria have no nuclei.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Chloroplast'.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Frequency of T is 0.47 and frequency of t is 0.53'.
9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'photosynthesis to produce glucose'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** The baby contracted it from a contaminated surface.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** Severing the dorsal root will lead to loss of sensation and reflex action, but if the ventral root is severed, the rat would sense the stimulation but would not be able to react.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key